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RESEARCH ARTICLE

Deep Learning-Based Multi-Floor Indoor Localization Using Smartphone IMU Sensors With 3D Location Initialization

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ABSTRACT Pedestrian Dead Reckoning (PDR) is a widely used indoor localization technique that does not require infrastructure and can exploit smartphone inertial sensors. However, it suffers from accumulated drift, and accurate initial position and floor information are especially important in multi-floor environments. This study proposes a deep learning-based multi-floor indoor localization method that estimates a three-dimensional position using only smartphone sensors without relying on external infrastructure. The proposed method utilizes the smartphone magnetometer to perform accurate 3D position initialization and drift correction, thereby addressing major limitations of conventional PDR methods such as sensitivity to initialization errors and performance degradation caused by drift. In this paper, we explain conventional IMU sensor-based PDR methods and describe the problems of PDR as well as the limitations of barometer-based floor transition detection techniques. The proposed initial 3D position estimation method detects the initial floor using a deep learning model trained on magnetic field sequences from the user's first 10 steps, and estimates the 2D position by comparing the current magnetometer data with a pre-built magnetic field map using the k-nearest neighbors algorithm. After detecting the initial 3D position, the method combines barometer-based floor transition detection and PDR-based position estimation to enable multi-floor indoor localization, and addresses the drift problem using correction nodes based on magnetic field data. As a result of experiments conducted in multi-floor buildings, the proposed method demonstrated significantly superior performance in terms of path similarity and localization accuracy compared to conventional methods based on PDR, complementary filter, and Kalman filter. In particular, it achieved a localization error of 64 cm and up to 99.8% accuracy within a 2-meter threshold, proving the effectiveness and practicality of the proposed approach.

INDEX TERMS Smartphone, multi-floor indoor localization, deep learning, pedestrian dead reckoning, inertial measurement unit, magnetic field map, 3D location initialization.

I. INTRODUCTION

In recent years, location-based services (LBS) using smartphones have attracted significant attention. Localization techniques can be broadly categorized into outdoor and indoor localization. In outdoor environments, the Global Navigation Satellite System (GNSS), which relies on satellite signals, is commonly employed to determine a user's current location. However, GNSS signals suffer from severe attenuation and multipath effects in indoor environments, which significantly degrade positioning reliability. This limitation has necessitated the development of alternative indoor localization technologies.

Indoor localization methods can be classified into two main categories: infrastructure-dependent and infrastructure-independent techniques [1]. Infrastructure-based methods estimate the user's location by leveraging the signal strength of wireless communication systems, such as Wi-Fi access points (APs) or Bluetooth Low Energy (BLE) beacons [2], [3], [4]. Despite their widespread use, these methods face

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challenges in accuracy due to environmental factors such as path loss, shadowing, and signal fading [5], [6]. Furthermore, in emergency situations such as natural disasters where wireless infrastructure may be damaged or unavailable, these methods become inapplicable. As a result, there is increasing demand for accurate indoor localization methods that do not rely on external infrastructure [7], [8].

Infrastructure-independent methods estimate a user's location by utilizing only the sensors embedded in smartphones. These approaches have recently gained attention due to their cost-efficiency and ease of deployment [9]. Among them, Pedestrian Dead Reckoning (PDR) is one of the most actively studied techniques. PDR uses smartphone sensors such as accelerometers, gyroscopes, and magnetometers, all of which are components of micro-electro-mechanical system (MEMS)-based inertial measurement unit (IMU). The accelerometer is used to estimate the user's stride length, while direction estimation is achieved either by combining the accelerometer with the magnetometer or by using the gyroscope alone. This enables estimation of the current location based on movement from a known previous position [10].

However, PDR has a critical limitation: its performance degrades over time due to gyroscope drift. Drift refers to the cumulative error that arises when integrating angular velocity measurements from the gyroscope, being a primary cause of reduced accuracy in indoor localization. To mitigate this issue, various noise reduction techniques, such as Kalman filters (KF) or complementary filters (CF), are commonly applied [11], [12]. Additionally, as PDR is a relative positioning method, uncertainty in the user's initial location can negatively affect the overall accuracy.

In multi-floor indoor environments, accurately identifying the user's floor is just as important as estimating their 2D position. Infrastructure-independent methods for multi-floor localization typically use barometric sensors embedded in smartphones to detect changes in air pressure between floors [13], [14]. However, these sensors are highly sensitive to atmospheric conditions such as temperature and weather, resulting in inconsistent readings even at the same location on different days. While this sensitivity is advantageous for detecting floor transitions, it limits the ability to accurately determine the user's initial floor [15].

To enhance the effectiveness of barometric sensor-based multi-floor indoor localization systems, robust initial floor detection is essential. This study aims to address this need through the following objectives:

- Detect the user's initial floor using a magnetic field map and deep learning models to satisfy the fundamental requirements of a barometric sensor-based multi-floor indoor localization system.
- Estimate the user's initial location using a magnetic field map after detecting the initial floor, thereby meeting the core requirements of the PDR system.
- Correct drift errors in the existing PDR system by applying calibration nodes (CN) utilizing magnetic field data when floor transition signals are detected, thereby ensuring more stable and accurate multi-floor indoor localization.

To accomplish these objectives, this paper proposes an infrastructure-independent multi-floor indoor localization method that integrates IMU sensors with magnetic field maps through deep learning and machine learning. Figure 1 illustrates an overview of the proposed approach.

In the proposed method, the smartphone's magnetometer is used to construct a magnetic field map, which serves as training data for a deep learning model. A long short-term memory (LSTM) network is utilized to detect the initial floor by analyzing sequences of magnetic field measurements collected during user movement. After identifying the floor, the user's 2D location is estimated using the k-nearest neighbor (k-NN) algorithm, which is applied to the magnetic field map of the detected floor. By combining the floor and horizontal location information, the system determines the user's initial 3D location, which serves as the initialization point for the subsequent localization process.

Next, barometric sensor data are used to detect inter-floor movements. A sequence-to-sequence (Seq2Seq) deep learning model is employed to analyze pressure changes between steps, enabling accurate floor transition detection. In this paper, the term "3D location initialization" specifically refers to determining the floor level and 2D horizontal position, rather than full 3D spatial coordinates.

This paper is organized as follows: Section II addresses the challenges of infrastructure-independent multi-floor indoor localization using smartphones. Section III proposes solutions to overcome these challenges. Section IV evaluates the performance of the proposed method through experiments and analysis. Finally, Section V concludes the paper with a summary and discusses future research directions.

II. EXISTING LOCALIZATION SYSTEMS

This section provides an overview of existing multi-floor indoor localization methods in infrastructure-free environments using smartphones. It also discusses the key challenges associated with these methods, including initial floor detection, initial location estimation, and drift issues.

A. INDOOR FLOOR DETECTION PROBLEM

In conventional multi-floor indoor localization systems, floor detection is typically achieved using wireless communication signal strengths, such as Wi-Fi and beacon signals. Access points installed indoors emit unique signal strengths, which vary depending on the user's location. Leveraging this characteristic, widely used fingerprinting techniques record Wi-Fi or BLE signal strength information on different floors and locations in a database. Real-time signal data are then compared with this database to estimate the user's 3D location [16], [17].

However, in infrastructure-independent environments, using wireless communication signals such as Wi-Fi and BLE is not feasible. As an alternative, floor detection often

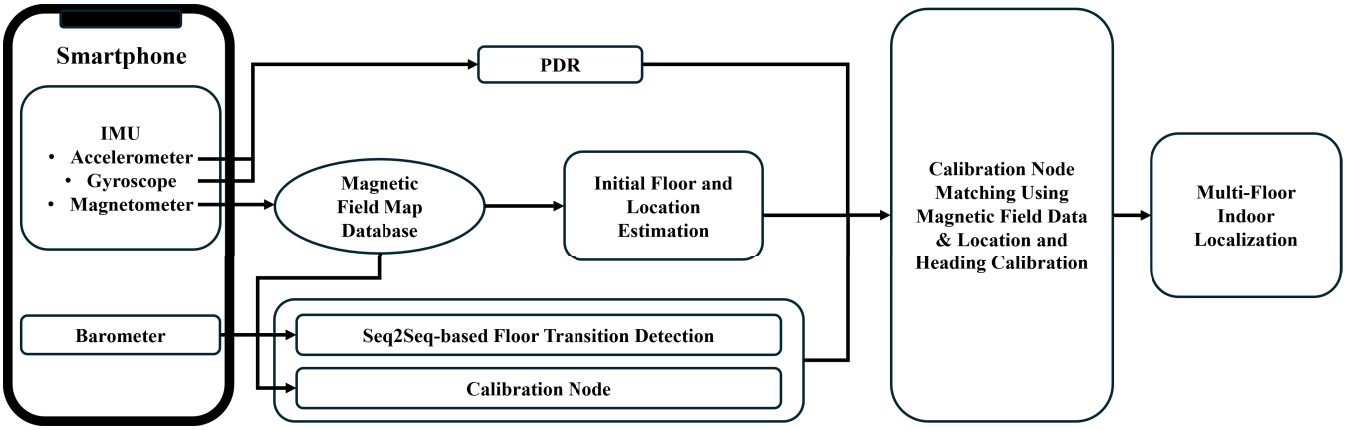

FIGURE 1. Architecture of our scheme.

relies on a barometric sensor embedded in the smartphone. A smartphone barometer measures the atmospheric pressure at the user's location enabling estimation of altitude. Barometric data are particularly useful for detecting floor changes caused by the use of staircases or elevators. By analyzing these pressure changes, the number of floors traversed by the user can be calculated.

The integration of barometric data with deep learning models has enhanced floor detection accuracy. Deep learning models can learn the pressure change patterns that occur during floor transitions, such as ascending stairs, descending stairs, or using an elevator. These trained models match real-time barometric data with pre-constructed pressure maps to estimate the user's floor [18]. Such methods are versatile and can operate in various environments without relying on external infrastructure.

Despite their advantages, existing barometer-based floor detection methods have several limitations:

1) **Sensitivity to weather and temperature:** Barometric sensors are highly sensitive to weather and temperature changes. Since atmospheric pressure fluctuates based on weather conditions, measurements can vary at the same location depending on the time of day or external environmental factors. These fluctuations make it challenging to accurately determine the initial floor, negatively impacting the overall performance of multi-floor localization systems [19], [20].
2) **Dependency on initial floor information:** Barometer-based multi-floor localization systems require prior knowledge of the user's initial floor. If the initial floor is incorrectly set, all subsequent floor detection results will include errors. This dependency significantly affects the reliability and accuracy of such system.

To address these challenges, more precise initial floor detection and barometric data correction techniques that can operate independently of infrastructure are required. These improvements are essential for enhancing the accuracy and robustness of multi-floor indoor localization systems in diverse environments.

B. INDOOR INITIAL LOCATION UNCERTAINTY PROBLEM

Indoor localization systems in environments with infrastructure limitations primarily rely on PDR techniques that use only IMU sensors embedded in smartphones. PDR is a relative tracking technology that estimates the user's current location based on the user's movement trajectory. It calculates the stride length using accelerometers, estimates the direction using gyroscopes or by fusing accelerometer and magnetometer data, and determines the user's current location relative to the previous location. PDR involves the following fundamental steps:

1) **Stride length calculation:** The accelerometer data are analyzed to detect periodic steps during user movement, which are then used to calculate the stride length. Since stride length varies among individuals, a dynamic stride estimation algorithm tailored to the walking patterns of the user is required [21].
2) **Direction estimation:** Direction estimation involves extracting rotational information from the gyroscope, which captures angular velocity changes during movement. The user's heading is derived from these measurements. For improved stability, gyroscope data can be complemented with magnetometer readings to mitigate drift and noise [22].
3) **Current location calculation:** The user's current location P_k at step k is estimated using the calculated stride length and direction. This can be expressed by the following equation:

$$P_k = P_{k-1} + \lambda_k \cdot \begin{bmatrix} \sin\theta_k \\ \cos\theta_k \end{bmatrix}. \tag{1}$$

Here, P_{k-1} is the previous location, λ is the stride length, and θ is the heading angle. This equation demonstrates that PDR is a relative localization method and necessitates accurate initial location information P_0.

Since PDR is inherently a relative method, inaccurate initial location information can result in significant cumulative errors in the entire localization process. Traditional infrastructure-based indoor localization systems address this

issue by using Wi-Fi or BLE signals or by manually specifying the initial location [23], [24]. However, these options are unavailable in infrastructure-free environments, leading to limitations in the initial location estimation.

To overcome these limitations, researchers have proposed leveraging a smartphone's magnetometer to build an indoor magnetic field map, which can be used to estimate the user's location [25]. Earth's magnetic field is distorted by structural elements, metal structures, and electronic devices within buildings, creating unique magnetic field characteristics for each indoor environment. These unique magnetic field patterns vary by location, enabling the creation of a database of magnetic field values for specific points within a building [26].

The constructed magnetic field map contains unique magnetic field vectors, each measured at a specific point and stored in a database. Deep learning models trained on this magnetic field map compare real-time magnetic field data against the database to predict the user's location. This magnetic field map-based approach eliminates the need for additional infrastructure and leverages existing magnetic field characteristics for localization.

Deep learning models, particularly recurrent neural networks (RNNs), are well-suited for analyzing time-series data patterns, such as those found in magnetic field sequences. An RNN processes sequential magnetic field data collected along the user's trajectory, enabling predictions of the user's movement path. The continuous variations in the magnetic field data represent specific paths, and the RNN effectively handles such time-dependent data [27], [28].

However, challenges remain in using RNN models for magnetic field-based localization. Since RNN models rely on accurate initial states, mismatches between the trained magnetic field map and real-time input data can significantly reduce the accuracy of the predictions. This mismatch may result in distorted trajectory predictions and degraded overall localization performance.

Accurately estimating the user's initial location is essential for achieving reliable indoor localization in magnetic field-based systems. Developing robust techniques to initialize the system with precise starting location information is critical for addressing these challenges and enhancing the overall performance of indoor localization systems in infrastructure-free environments.

C. PDR DRIFT PROBLEM

The gyroscope, an internal IMU sensor in a smartphone, is a critical sensor for calculating orientation. It utilizes the Coriolis effect to measure angular velocity around the X, Y, and Z-axes in units of rad/sec. The rate of change of Euler angles varies depending on the relative angular velocity between different body axes. The state transformation of the IMU to calculate the rate of change in the inertial frame is governed by the following transformation process [29], [30].

$$F_c = -2m(\omega * v), \tag{2}$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi)\tan(\theta) \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}. \tag{3}$$

(2) describes the Coriolis effect, representing the force F_c that acts perpendicular to both the axis of rotation and the velocity of a mass m moving at velocity v in a rotating reference frame with angular velocity ω. This force is measured in MEMS gyroscopes to estimate angular velocity. In (3), ω_x, ω_y, and ω_z are the angular velocities measured by the gyroscope in the body frame along the X, Y, and Z axes, respectively.

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \int_0^t \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} dt + \begin{bmatrix} \phi_0 \\ \theta_0 \\ \psi_0 \end{bmatrix}. \tag{4}$$

Euler angles are obtained by integrating angular velocity measurements from the gyroscope over time, as shown in Equation (4). Here, ϕ_0, θ_0, and ψ_0 denote the initial roll, pitch, and yaw angles at the start of the measurement.

However, Euler angles are susceptible to a phenomenon known as gimbal lock, which occurs when two of the three rotational axes align at certain angles, reducing the degrees of freedom for rotation and making it difficult or impossible to achieve the desired rotation [31], [32]. This phenomenon typically occurs when the pitch angle approaches ±90 degrees. As the pitch angle nears 90 degrees, yaw and roll rotations become indistinguishable, preventing the sensor from accurately estimating orientation using Euler angles.

To address this issue, quaternions are used to represent rotation. Quaternions represent rotations in 3D space as unit quaternions in 4D space, completely avoiding the gimbal lock problem associated with Euler angles. Quaternions enable continuous and singularity-free rotation representation in all directions, without axis overlap. Additionally, quaternions integrate the angle and axis of rotation into a single data structure, offering higher computational efficiency and numerical stability compared to rotation matrices or Euler angles.

Since quaternions do not require explicitly defining rotation axes or maintaining rotation order, they are highly reliable in environments with frequent multi-axis rotations. The process of calculating angles using quaternions follows the steps outlined below [33], [34], [35].

$$\omega = \begin{bmatrix} 0 \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}^T, \tag{5}$$

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \omega, \tag{6}$$

$$\dot{\mathbf{q}}_t = \frac{1}{2}\mathbf{q}_{t-1} \otimes \omega_t, \tag{7}$$

$$\mathbf{q}_{t+\Delta} = \mathbf{q}_{t-1} + \dot{\mathbf{q}}\Delta t. \tag{8}$$

In (5), ω represents the angular velocity vector in quaternion form, where ω_x, ω_y, and ω_z correspond to the angular velocities around the X, Y, and Z axes of the gyroscope

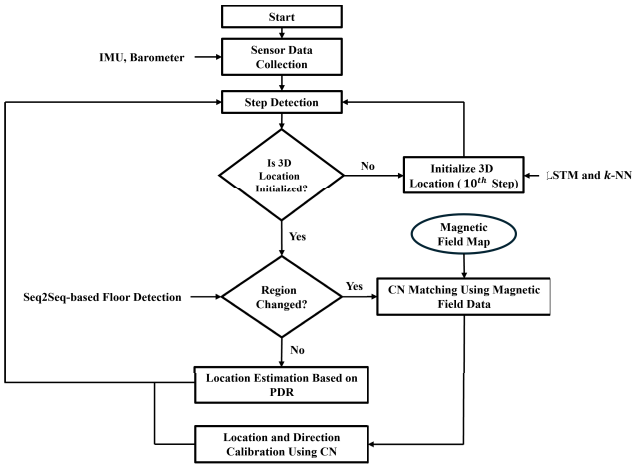

FIGURE 2. Flowchart of the proposed method.

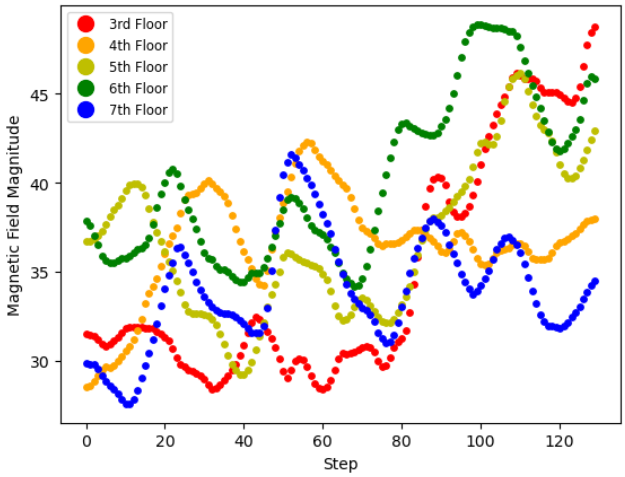

FIGURE 3. Magnetic field data sequences for each floor.

sensor frame, respectively. In (6), $\dot{\mathbf{q}}$ represents the quaternion rate of change of the Earth frame relative to the sensor frame, and $\mathbf{q}$ denotes the quaternion representing the rotation from the sensor frame to the Earth frame. The symbol $\otimes$ represents the quaternion multiplication. In (7), $\dot{\mathbf{q}}_t$ refers to the quaternion rate of change at time t, and $\mathbf{q}_{t-1}$ is the previously estimated quaternion at the previous time step. In (8), Δt is the sampling period, and $\mathbf{q}_{t+\Delta t}$ is the current quaternion computed by integrating the quaternion rate of change $\dot{\mathbf{q}}_t$ over the sampling period starting from the previous quaternion $\mathbf{q}_{t-1}$. This process enables continuous estimation of orientation from the sensor frame to the Earth frame.

In direction estimation techniques using gyroscope sensors, drift phenomena occur where cumulative errors arise during the integration of angular velocity, leading to increasingly inaccurate direction estimation over time [36]. Gyroscope drift is a critical issue in indoor localization and in various sensor fusion-based systems. If drift increases, the direction of user movement may be calculated incorrectly, and in particular, inaccuracies in direction information can cause significant errors in final location estimation in smartphone-based PDR techniques. To mitigate such drift, various correction methods have been proposed.

First, a method combines the absolute directional information from a magnetometer with posture estimation information from an accelerometer. Algorithms such as KF or CF can then be used to fuse this data with gyroscope measurements to correct drift [12]. Second, periodic reinitialization methods reset gyroscope values under specific conditions to reduce the accumulation of drift [37]. Third, software CN can be set up within a building to periodically recalibrate location and direction [18]. However, even these correction methods cannot fundamentally overcome the hardware limitations.

In indoor environments, challenges such as magnetic sensor distortion caused by steel structures in buildings and electromagnetic interference from electronic devices make it difficult to obtain accurate absolute directional measurements. Therefore, addressing the drift issue requires not only the development of new algorithms but also additional approaches to overcome the limitations of existing correction methods.

III. PROPOSED INDOOR LOCALIZATION METHOD

The proposed indoor localization method operates as illustrated in Figure 2, and is designed to address various issues associated with conventional multi-floor indoor localization methods in infrastructure-free environments, as described in Section II. This section first explains the 3D location initialization technique and then presents an effective method to resolve drift problems that may arise during the location estimation process.

A. 3D LOCALIZATION INITIALIZATION METHOD

This subsection details the procedures for floor detection and initial location estimation to initialize 3D localization. These procedures aim to address the challenges in multi-floor indoor localization discussed in II-A and II-B. The results of this subsection establish the conditions necessary for the effective operation of PDR-based and barometer-assisted multi-floor indoor localization methods.

1) MAGNETIC FIELD MAP BASED INDOOR INITIAL FLOOR DETECTION

The magnetometer, one of the IMU sensors embedded in smartphones, measures the strength and direction of the magnetic field in the surrounding environment. The magnetometer operates based on Earth's magnetic field and functions as a compass, pointing to the magnetic north of Earth [38]. However, in indoor environments, external factors, such as the steel structure of the building or electronic devices, distort the magnetometer readings. Due to this distortion, the accuracy of magnetometers is insufficient for standalone direction estimation, and they are primarily used as supplementary components in systems.

This paper proposes an initial floor detection method using Earth's indoor magnetic field data to address the challenges in barometer-based floor detection methods. The magnetic

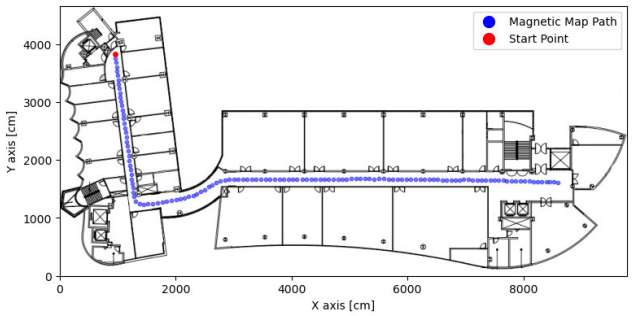

FIGURE 4. Magnetic field data collection path.

field sequence of each floor exhibits unique patterns due to the steel structure of the building and external factors, as illustrated in Figure 3. These patterns enable the detection of initial floor information. To demonstrate this approach, magnetic field data were collected from the 3rd to the 7th floors of Hyeongnam Engineering Building at Soongsil University using a Samsung Galaxy Note 10+. The data were collected using a custom-developed Android smartphone application, triggered at each step event. Data collection was carried out along the path shown in Figure 4, with the smartphone facing upward during walking. The structure of the building was identical on all floors. Magnetic field data were collected at each step and the number of steps per floor was maintained at 130.

Step detection was conducted using the accelerometer, which is one of the smartphone's IMU sensors. The accelerometer-based step detection process is described below [39].

$$\mathbf{a}_t^G = \mathbf{R}_t \mathbf{a}_t^L. \tag{9}$$

Here, a_t^G represents the acceleration vector transformed into the global coordinate system, while a_t^L represents the acceleration vector in the local coordinate system. Additionally, R_t denotes the rotation matrix, which was calculated using the `SensorManager.getRotationMatrix()` method based on the values from the accelerometer and magnetometer sensors.

$$\{a_t^z > acc^{upper},\ a_{t+\Delta t}^z < acc^{lower},\ 0 < \Delta t < 1\}. \tag{10}$$

Here, a_t^z represents the vertical acceleration, acc^{upper} and acc^{lower} are thresholds set for step detection, and Δt denotes the time interval between the maximum and minimum accelerations. The magnetometer values are expressed as 3D vectors, which change depending on the smartphone's orientation. As a result, even at the same location, the magnetometer readings may differ depending on the smartphone's orientation. This orientation dependency of the magnetometer can cause errors in magnetic field mapping, as the map may fail to accurately capture location-specific characteristics. To address this issue, this study aligns all collected magnetic field vectors to a reference direction before constructing the magnetic field map. The reference direction was set to the north on the map, corresponding to 0 degrees, as shown in Figure 4. During experiments,

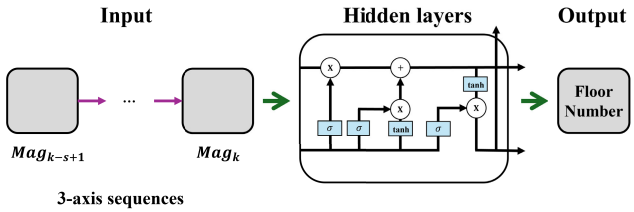

FIGURE 5. LSTM model overview.

all magnetic field vectors were aligned to this reference direction, ensuring that the characteristics of a given location could be accurately identified regardless of the smartphone's orientation. The rotational transformation of the magnetic field vectors was performed through the following process.

$$\mathbf{V}_{mag} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \tag{11}$$

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{12}$$

$$\begin{aligned} \mathbf{V}'_{mag} &= \mathbf{R}_z(\theta)\mathbf{V}_{mag} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta \cdot v_x - \sin\theta \cdot v_y \\ \sin\theta \cdot v_x + \cos\theta \cdot v_y \\ v_z. \end{bmatrix} \end{aligned} \tag{13}$$

Here, $\mathbf{V_{mag}}$ represents the magnetic field vector, and v_x, v_y, v_z are its components. The smartphone's rotation angle from the current reference direction is denoted as θ, which is calculated using the embedded gyroscope sensor. The rotation matrix for axis rotation is denoted as $\mathbf{R}$, and the rotated magnetic field vector $\mathbf{V}'_{mag}$ is calculated using the magnetic field vector and rotation matrix obtained in (11) and (12), following the relationship in (13). Data augmentation techniques are often used to enhance the generalization performance of models or prevent overfitting when training data is limited. In this study, magnetic field sequence data were collected for 130 steps per floor. To increase data diversity without conducting additional collection, data augmentation was performed by applying rotation transformations to the magnetic field sequences from each floor. Random angles within a range of ± 10 degrees were applied to the magnetic field sequence data for each floor, generating 60 magnetic field sequences per floor. Of these, 30 were reversed sequences, starting from the end of the originals, enabling the deep learning model to also learn magnetic field patterns along reverse paths. This augmentation process diversified the magnetic field data for each floor, expanding the training dataset to improve model generalization and prevent overfitting.

The sliding window technique, a useful method for processing time-series data, was applied to the augmented data. A fixed-size window slides over the data, with the data

TABLE 1. Hyperparameters used in LSTM model training.

Model parameters	Values
Number of LSTM layers	2
Parameters in LSTM layers	116,992
Number of trainable parameters	119,237
Activation function of hidden layers	ReLU
Activation function of output layer	Softmax
Initializer	Xavier uniform
Loss function	Categorical cross entropy
Optimizer	Adam
Learning rate	0.001
Number of epochs	30

within each window used for model training [40]. In this study, a window size of 10 was used, transforming the original 650×3 training dataset into a 36,300×10×3 dataset through data augmentation and sliding window techniques. The transformed dataset was divided into training, validation, and test sets in an 8:1:1 ratio, resulting in 29,040, 3,630, and 3,630 samples, respectively.

Figure 5 illustrates an overview of the proposed LSTM model. LSTM networks are highly effective for processing time-series data and capturing temporal patterns, particularly in indoor localization tasks [41]. The LSTM model was trained offline using preprocessed 3-axis magnetic field data as input and corresponding floor information as labels. In the online phase, the LSTM model received the same preprocessed 3-axis magnetic field data as input and predicted the user's floor in real time.

Key training hyperparameters are summarized in Table 1. The rectified linear unit (ReLU) activation function was used within the hidden layers to mitigate the vanishing gradient problem and accelerate training. The Softmax activation function was applied at the output layer to enable multi-class floor classification. Categorical cross-entropy was used as the loss function to optimize the difference between the predicted probabilities and the true floor labels. The Adam optimizer was employed to ensure stable and efficient convergence during training.

2) MAGNETIC FIELD MAP BASED INDOOR INITIAL LOCATION ESTIMATION

After obtaining the floor information through the initial floor detection process introduced in Section III-A-I, adding only the 2D initial location completes the requirements for implementing a multi-floor indoor localization method in an infrastructure-limited environment. To meet these conditions, we estimate the user's initial location by applying the k-NN algorithm to a pre-built magnetic field map. The k-NN algorithm is a simple and intuitive algorithm widely used for classification and regression in supervised learning. It predicts the class or value of a new data point by identifying the k-nearest neighbors among existing data and making predictions based on their information [42].

The k-NN algorithm typically uses Euclidean distance to calculate the proximity between neighbors, although other distance metrics such as Manhattan distance or cosine similarity can also be applied. In this study, we selected Euclidean distance to measure the similarity between magnetic field data points. The Euclidean distance between two data points x and y with n dimensions is calculated as follows:

$$\text{Distance}(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}. \tag{14}$$

Based on this distance, the algorithm selects the k-nearest neighbors to the new data point. For classification problems, the new data point is assigned the class most frequent among the k neighbors. For regression problems, the predicted value is calculated as the average of the values of the k neighbors. In this study, we approach the problem as a regression task using the k-NN algorithm. Specifically, a pre-constructed magnetic field map of each floor in the building serves as the database, and the 3-axis Earth's magnetic field values collected via the smartphone's magnetometer are used as input data. The magnetometer data are represented as 3D vectors, and the corresponding label data are the location coordinates on the building's blueprint.

The algorithm operates after the initial floor information is determined. When the smartphone user launches the application and takes 10 steps, a magnetic field sequence dataset for those steps is generated. These data are used for initial floor detection, and the magnetic field data from the 10th step is compared against the magnetic field map of the detected floor. The k-NN algorithm then predicts the user's 2D location. Through this process, the user's 3D location can be estimated after just the first 10 steps, regardless of their location indoors. This approach effectively eliminates the limitations associated with using barometer-based floor transition detection and PDR techniques with IMU sensors.

B. 3D LOCATION TRACKING METHOD

This subsection introduces a method for correcting drift in the PDR technique using smartphone IMU sensors, as discussed in Section II-C, by employing CN. After initializing the user's 3D location, the proposed method utilizes the barometer sensor to detect floor transitions and prevents drift by applying CN. This process can be considered a form of magnetic fingerprinting, where location estimation is based on matching measured magnetic field patterns to a pre-constructed database.

1) FLOOR TRANSITION DETECTION USING BAROMETER SENSOR

A barometer is a sensor that measures atmospheric pressure and is widely used in devices, including smartphones, to detect changes in altitude. Since atmospheric pressure varies with altitude, barometer data can be used to estimate

TABLE 2. Hyperparameters used in Seq2Seq model training.

Model parameters	Values
Model type	Seq2Seq (Encoder–Decoder)
Encoder LSTM units	16 (1,152 params)
Decoder LSTM units	16 (1,152 params)
Number of trainable parameters	2,355
Activation function of hidden layers	tanh
Activation function of output layer	Softmax
Initializer	Xavier uniform
Loss function	Sparse categorical cross entropy
Optimizer	Adam
Learning rate	0.001
Number of epochs	30

the relative altitude of a smartphone user. This provides essential information for implementing multi-floor indoor localization methods in infrastructure-free environments. In this study, we combined barometric sensor data with a deep learning model to develop a multi-floor indoor localization method capable of accurately detecting floor transitions [14], [18].

To implement the floor transition detection method, data were collected for floor transitions occurring on flat surfaces and stairs at Hyeongnam Engineering Building of Soongsil University. Barometer sensor data were stored in the smartphone database only when steps were detected. The collected barometer data were preprocessed using mean normalization to remove the average value. Additionally, a sliding window technique was applied to the data to help the deep learning model effectively learn temporal patterns. The window size was set to 15.

In this study, a Seq2Seq deep learning model was used to process the time-series barometer sensor data. The Seq2Seq model consists of an encoder and a decoder, typically based on RNN such as LSTM [43]. The encoder compresses the input sequence into a fixed-size vector, and the decoder generates the output sequence based on this vector. The Seq2Seq model is effective for learning temporal patterns and handling cases where the input and output sequences differ in length. Table 2 lists the main hyperparameters used to train the Seq2Seq model.

This study framed the problem as a single-output classification task, where the Seq2Seq model classified each data point into one of three classes: 0 (Down), 1 (Flat), and 2 (Up). The model takes 15 consecutive changes in barometric pressure as input and predicts the user's current step state. After the user starts the application, the barometric differences for each step are stored in a queue of length 15. Small barometric changes are classified as Flat, while large changes are classified as Up or Down. Elevator-based floor transitions were identified using a threshold for barometric changes. The barometer measures pressure in millibars (mbar), and an empirical threshold of 0.3 mbar was set for a single floor transition. If the model predicts an Up or Down state and the pressure difference between consecutive steps exceeds 0.3 mbar, the user is considered to have used the elevator.

The predicted step states are combined with a pre-built relative pressure map for further use. In this study, a relative pressure map of the entire building was constructed to enable accurate floor number detection during transitions. The relative pressure map was built using the following steps:

- **Pressure data collection:** Atmospheric pressure data were collected using a barometer sensor on each floor, and the average pressure value for each floor was calculated.
- **Pressure-to-altitude conversion:** Atmospheric pressure data were converted into altitude values (in meters) to calculate the height of each floor. The conversion process is as follows [44]:

$$h_i = 44330 \times \left(1 - \sqrt[5.255]{\frac{p_i}{p_0}}\right). \tag{15}$$

  Here, h_i represents the relative height, while p_i and p_o denote the barometer reading (atmospheric pressure) and the standard atmospheric pressure at sea level (in mbar), respectively.
- **Relative altitude calculation:** Based on the converted altitude data, the height difference between floors was calculated and normalized to determine the relative floor heights.
- **Relative pressure map construction:** The altitude values for each floor were stored in a dictionary format to create a relative pressure map. This map matches the user's current floor to the nearest floor based on altitude changes, estimating the new floor number.

The relative pressure map supports accurate floor transition detection in multi-floor indoor environments and enables real-time verification of the user's current floor.

2) MAGNETIC FIELD MAP BASED LOCATION CALIBRATION

The PDR technique is a useful method for estimating a user's location and heading during movement. However, as previously mentioned, PDR-based systems suffer from drift, which gradually degrades localization accuracy over time. This study proposes a method to mitigate drift by utilizing CN based on magnetic field data. Unlike conventional CN, which select the nearest node based on barometric pressure changes [14], [18], this study introduces a novel approach that leverages magnetic field data, enabling precise location correction using non-uniform magnetic field distributions. Additionally, the CN used in this study were implemented in software.

To implement this approach, magnetic field data were collected in advance near vertical transition areas, such as staircases and elevators, on each floor. Figure 6 illustrates the list of implemented CN. The proposed CN differ from conventional methods by integrating not only magnetic field

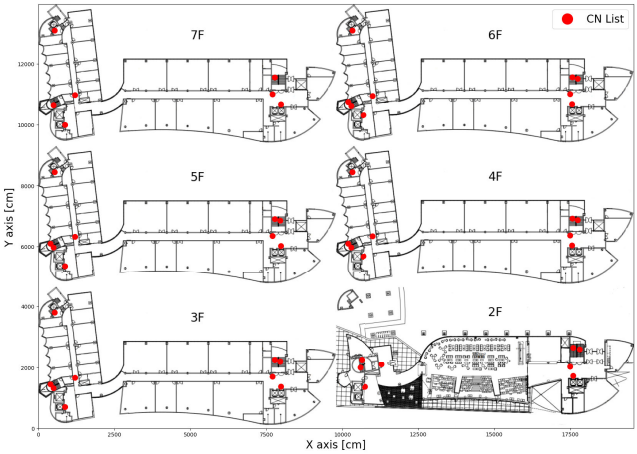

FIGURE 6. Calibration nodes at each floor.

data but also information about heading, location, floor level, and vertical transition mechanisms. This enables real-time and highly precise correction of the user's location and heading. The CN operate as follows:

1) **CN search and matching:** When the floor transition detection model identifies a floor change along the user's path, the current magnetic field data is compared with the predefined list of CN for the current floor. The system searches for and matches the CN with the most similar magnetic field data to the user's current location.
2) **Similarity calculation:** The similarity between the CN and the user's magnetic field data is calculated using the Euclidean distance of the 3-axis magnetic field data. The CN with the smallest similarity distance is selected, enhancing the reliability of the correction process.
3) **Calibration process:** The user's estimated location and direction are updated using the precise location and directional information provided by the CN. This effectively mitigates drift in the PDR technique and maintains high localization accuracy even over long travel distances. Consequently, users can achieve stable and reliable localization performance even in complex indoor environments or during extended movement.

CN-based updates to location and direction operate in a complementary manner with the PDR technique. When no floor transition is detected, the user's location is estimated using stride length and directional changes. Conversely, when a floor transition is detected and a correction is applied, the user's location and direction are updated using the information provided by the CN. In this study, the correction process was repeatedly applied based on the output of the floor transition detection model, effectively eliminating cumulative errors.

IV. EXPERIMENTAL RESULTS

This section presents the experiments and results of the deep learning and sensor fusion-based multi-floor indoor localization method. All experiments were conducted on the same device used for data collection: a Samsung Galaxy Note 10+. The section is organized into the following parts: experiments and results of initial floor detection for smartphone users, experiments and results of initial location estimation when floor information is known, and experiments and results of drift mitigation using CN after 3D location initialization.

TABLE 3. Initial floor detection accuracy.

Accuracy	3rd	4th	5th	6th	7th	Average
Hyeongnam	95%	86.8%	94.2%	95%	89.3%	92.06%
Cho Man-sik	95%	94.2%	97.5%	90.9%	90.9%	93.7%

TABLE 4. Confusion matrix for initial floor detection in Hyeongnam Engineering Building.

		Predicted class					
		3rd	4th	5th	6th	7th
Ground truth	3rd	**95.0%**	1.7%	0%	2.5%	0.8%
	4th	1.7%	**86.8%**	5.8%	2.5%	3.2%
	5th	1.7%	0%	**94.2%**	2.5%	1.6%
	6th	0%	1.7%	2.5%	**95.0%**	0.8%
	7th	0.8%	6.5%	1.7%	1.7%	**89.3%**

TABLE 5. Confusion matrix for initial floor detection in Cho Man-sik Memorial Hall.

		Predicted class					
		3rd	4th	5th	6th	7th
Ground truth	3rd	**95.0%**	0%	0%	4.1%	0.8%
	4th	0.8%	**94.2%**	4.1%	0%	0.8%
	5th	0.8%	0.8%	**97.5%**	0%	0.8%
	6th	0.8%	0%	1.7%	**90.9%**	6.6%
	7th	0.8%	0.8%	2.5%	5.0%	**90.9%**

A. INITIAL FLOOR DETECTION EXPERIMENT

In this study, initial floor detection experiments were conducted on the 3rd to 7th floors of Hyeongnam Engineering Building and Cho Man-Sik Memorial Hall at Soongsil University. To verify the validity of the proposed method, data collection was performed along paths opposite to those used for constructing the magnetic field map. The collected magnetic field data were processed using a sliding window technique, with a window size of 10, consistent with the preprocessing of the training data. Through this process, the data were transformed into a $121 \times 10 \times 3$ format and used as input to the model. As shown in Tables 3 to 5, the proposed magnetic map-based deep learning model demonstrated high floor detection performance across multiple test environments. Floor detection accuracy in Table 3 was calculated by applying a sliding window of size 10 to the magnetic

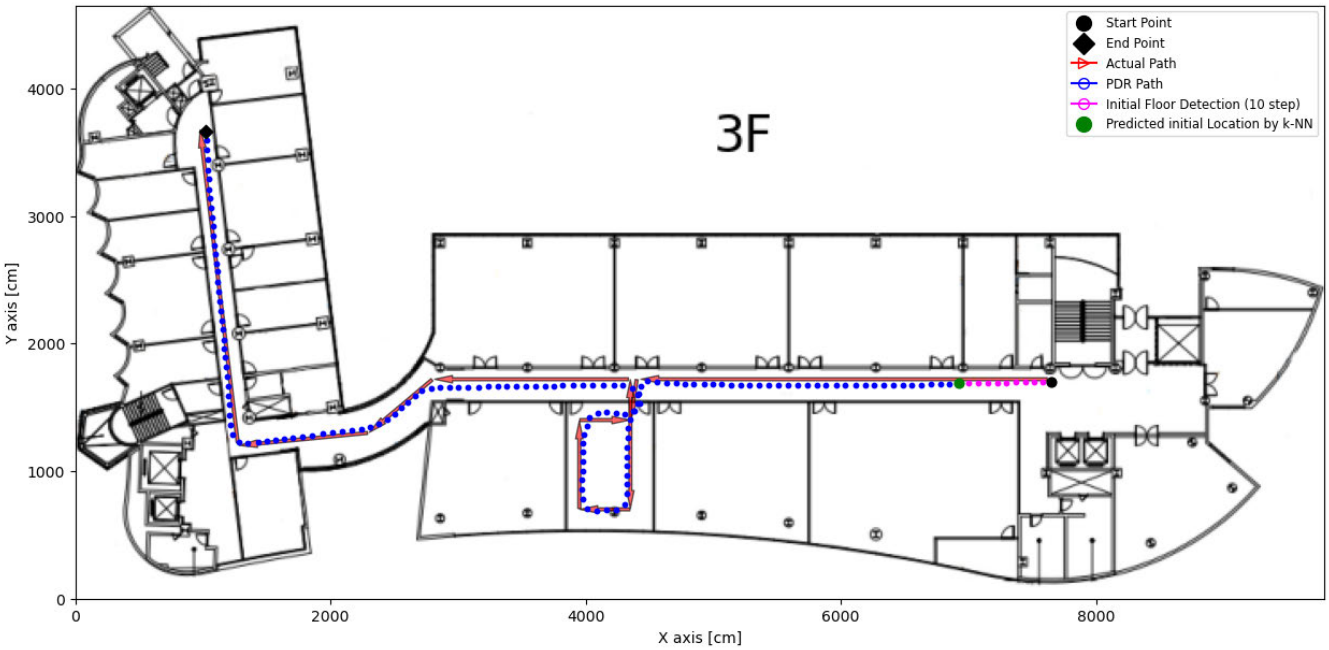

FIGURE 7. An example 3D location initialization experimental result in Hyeongnam Engineering Building (3rd Floor).

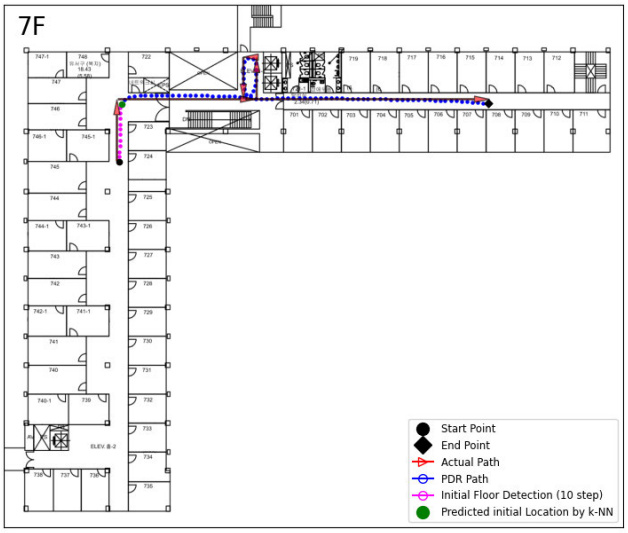

FIGURE 8. An example 3D location initialization experimental result in Cho Man-sik Memorial Hall (7th Floor).

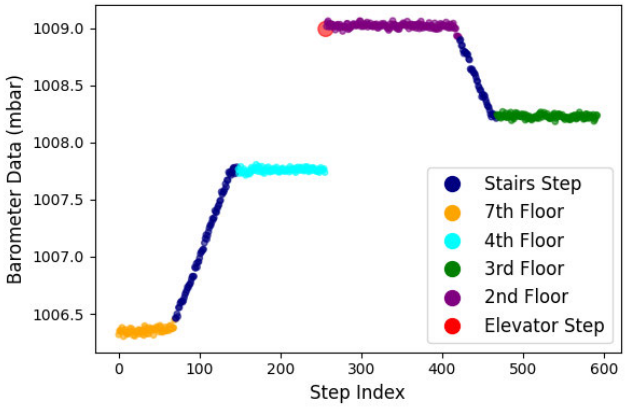

FIGURE 9. Floor transition experimental result.

field sequence. For each window, the model predicted the floor label, and if the predicted label matched the ground-truth floor, the prediction was counted as correct. The final accuracy was computed as the ratio of correct predictions to total windows. Table 3 presents the overall floor detection accuracy for each floor in both buildings: Hyeongnam Engineering Building and Cho Man-sik Memorial Hall. The average floor detection accuracy was 93.7% for Cho Man-sik and 92.06% for Hyeongnam, demonstrating robust generalization even across different buildings. Tables 4 and 5 provide detailed confusion matrices showing how the model classified each floor. These results validate the model's ability to learn distinct floor-level magnetic field patterns and satisfy the requirement for infrastructure-independent initial floor detection within the first 10 steps of user movement.

B. INITIAL LOCATION ESTIMATION EXPERIMENT

By determining the user's floor information through the process mentioned above, the PDR technique can be applied once the 2D location information is provided. In this study, the 2D location was estimated using the k-NN algorithm, with the k-value experimentally set to 3. Figure 7 presents the experimental results of combining initial floor detection and 2D location estimation conducted on the 3rd floor of Hyeongnam Engineering Building at Soongsil University.

As shown in Figure 7, the floor information was obtained from the magnetic field data of the first 10 steps.

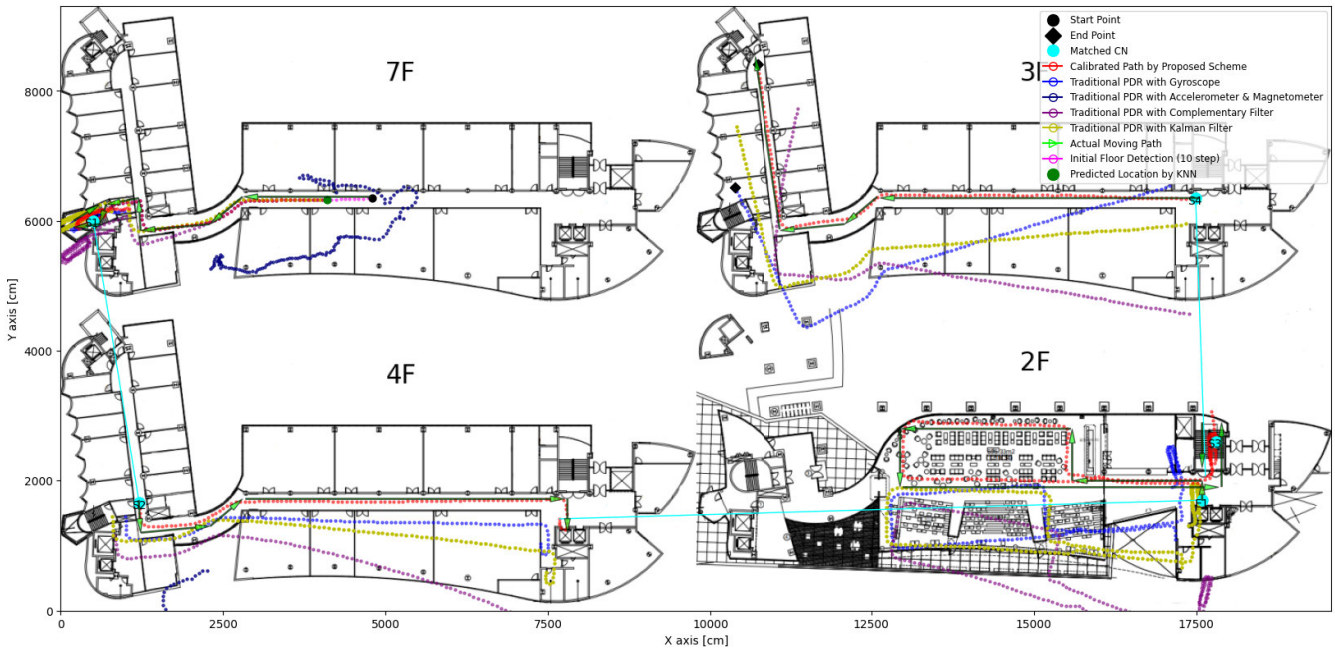

FIGURE 10. Experimental results of the proposed multi-floor indoor localization method.

The user's current location was then estimated by comparing the magnetometer data with the magnetic field map of the identified floor using the k-NN algorithm. Determining the initial floor and location satisfied the conditions for applying the PDR and barometer-based floor transition detection techniques. Subsequently, the PDR-derived path, represented by the blue trajectory, closely resembled the actual path, illustrating consistent and high-accuracy localization.

To further verify the effectiveness of the proposed method, the same experiment was conducted in Cho Man-sik Memorial Hall using the same procedure. The results are presented in Figure 8.

C. INDOOR MULTI-FLOOR LOCALIZATION IN INFRASTRUCTURE-FREE ENVIRONMENT

After estimating the user's 3D position using a smartphone, we conducted multi-floor indoor localization experiments. The experiments took place on the 7th, 4th, 3rd, and 2nd floors of Hyeongnam Engineering Building at Soongsil University. The 7th, 4th, and 3rd floors consisted of corridor-shaped spaces, while the 2nd floor was structured as a large lecture hall. The total walking distance during the experiment was approximately 600 meters.

The results of an example floor transition experiment are shown in Figure 9 where the pedestrian starts the traverse on the 7th floor and takes stairs to get to the 4th floor. After walking on the 4th floor for some time, takes the elevator to the 2nd floor. Subsequently, walks for some distance on the 2nd floor and takes the stairs again but this time arrives at the 3rd floor where the experiment is terminated after a short walk. The experimental results confirmed that upward steps, downward steps, stair usage, and elevator transitions were all accurately predicted. Figure 10 illustrates the localization results along the experimental path. The black point on the 7th floor indicates the starting point, and the pink points represent the first 10 steps used for initial floor detection. After floor detection, the green point indicates the estimated initial position.

TABLE 6. AR and RMSE evaluation results.

	AR $\epsilon = 0.5$	AR $\epsilon = 1.0$	AR $\epsilon = 1.5$	AR $\epsilon = 2.0$	RMSE [m]
Proposed scheme	48.9%	93.42%	98.65%	99.83%	0.64
PDR with Gyroscope	6.57%	11.8%	19.73%	26.3%	4.96
PDR with Acc&Mag	0.17%	0.17%	0.34%	0.34%	66.38
PDR with CF	0.34%	3.04%	6.58%	7.59%	8.4
PDR with KF	0.34%	3.04%	8.6%	10.79%	6.12

Matched CN are marked in light blue. For example, S1 represents the action of descending stairs from the 7th to the 4th floor, detected and matched by the floor transition model. S2 corresponds to the action of entering the 4th floor via stairs, while E1 represents the action of moving from the 4th to the 2nd floor using an elevator. S3 and S4 are also matched CN identified through step state change detection.

The actual path is shown with green arrows, and the localization results of the proposed method are represented

in red. The paths from the uncorrected PDR techniques are shown in blue (gyroscope only), navy (accelerometer and magnetometer), purple (complementary filter), and yellow (Kalman filter), respectively. Root mean square error (RMSE) and accuracy rate (AR) were used as performance metrics to evaluate the accuracy of the multi-floor indoor localization method. The calculated performance metrics are presented in Table 6, and AR and RMSE were calculated through the following process [45].

$$RMSE(P_k, \hat{P}_k) = \sqrt{\frac{\sum_{k=1}^{N}(P_k - \hat{P}_k)^2}{N}}, \tag{16}$$

$$\delta_k = \sqrt{(P_k - \hat{P}_k)^2}, \quad 0 < k \leq N, \tag{17}$$

$$AR = \frac{\#\{\delta_k \mid \delta_k \leq \epsilon\}}{N} \cdot 100\%. \tag{18}$$

RMSE represents the root mean square error between the actual location P_k and the predicted location $\hat{P}_k$, while AR measures the percentage of predicted positions within a predefined error boundary ϵ around the actual location. Experimental results revealed that conventional multi-floor indoor localization methods have limitations, such as their reliance on infrastructure for initial location estimation and reduced accuracy caused by drift. In contrast, the proposed method estimated the user's initial location without relying on infrastructure and achieved a position closely matching the actual location. Furthermore, by utilizing CN, it significantly reduced localization errors caused by drift during floor transitions. These results demonstrated stable and high localization performance, closely resembling the actual path.

V. CONCLUSION AND DISCUSSION

This paper proposed a novel approach to address the challenges of 3D location initialization and drift-induced localization errors in multi-floor indoor localization systems using only smartphone sensors. By leveraging a magnetic field map constructed from the smartphone's magnetometer and a deep learning model, the system enables accurate 3D location initialization using the user's first 10 steps, without any external infrastructure. Furthermore, the introduction of CN based on magnetic field data effectively mitigates drift errors in conventional PDR methods, maximizing the advantages of barometric sensors.

Experimental results demonstrated that the proposed method achieves high localization accuracy, closely matching ground-truth trajectories and significantly outperforming conventional PDR techniques. These findings validate the feasibility of achieving robust, infrastructure-independent multi-floor indoor localization solely with smartphone sensors. The major contributions of this study include (1) the development of a deep learning-based method for 3D initial location estimation using magnetic field sequences, and (2) the introduction of a drift correction technique utilizing magnetic field-based CN without requiring any external infrastructure.

Future research will focus on improving pose-invariant localization performance by enhancing the model's robustness to various smartphone carrying positions and orientations. Specifically, we aim to incorporate sensor fusion and orientation normalization techniques to minimize performance degradation caused by changes in device placement, such as in the hand, pocket, or backpack. Several studies have explored pose-invariant inertial navigation using deep learning techniques for smartphones [46], [47]. Building upon these insights, we plan to extend the proposed method's adaptability to dynamic user contexts and broader device heterogeneity. Through these efforts, we aim to develop a more generalizable and reliable infrastructure-free smartphone-based indoor localization system capable of maintaining high accuracy regardless of user pose or device orientation.

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